



Freeing the Trapped Lung

Restoring re-expansion when an organized rind traps the lung

Clinical Need

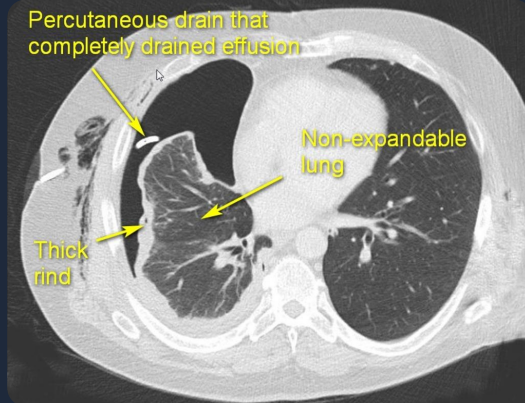
A way to **restore lung re-expansion** for **patients with a thick pleural rind** in order to **preserve lung function**, avoiding both decortication morbidity and trapped lung disability.

U.S. Population

1 million pneumonia patients:

→ **20-40%** develop parapneumonic effusion

→ **5-10%** develop empyema (~32k patients/yr)³



Trapped lung: rind blocks re-expansion

Measurable Outcome

Forced Vital Capacity²
60% → 80%

post decortication.

Imaging-confirmed re-expansion in 3-6 months.

In empyema patients alone:

20%

1-year mortality Rate
(35% in frail/elderly)⁴

~14 Days

Average Stay¹

~\$38.6K

Cost / Admission¹

30%

require surgery³

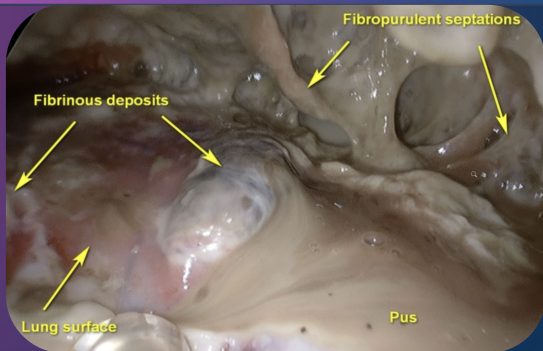
37% Rising

Incidence Rate
(2007-2016)¹

¹ Mummadi et al., 2021, Chest, 160(5):1534-1551 | ² Thoracic Research and Practice, 2019

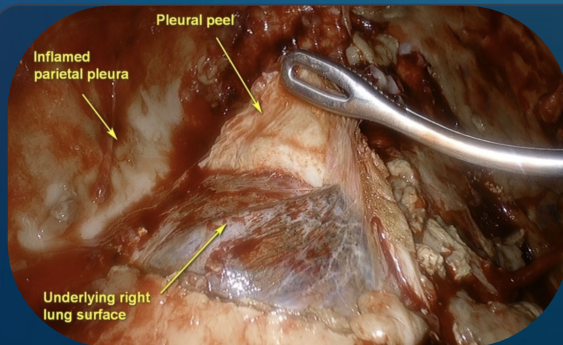
³ Shen et al., 2017 | ⁴ ERS/ESTS statement, 2022

The Surgical Dilemma



Stage II Empyema

- Loculated pus and fibrin.
- Pulmonologists drain and instill lytics here.
- tPA/DNase break up fibrin.
- **Works ~50-78% of the time**
(CTICU observed vs. published⁷)



Stage III Empyema

- Weeks later, fibrin matures into a collagen rind.
- Pulmonology to thoracic surgery
- ~30% of empyema patients end up here

Operable

BRUTAL DECORTICATION

- Air leak & bleeding
- Months to recover

Inoperable

ABANDONED PATHWAY

- Chronic breathlessness
- Cycles back for drainage

The longer the rind organizes, the worse both paths get — delay compounds surgical morbidity⁵.

Why the Rind Requires a Knife

01 / INCEPTION

Undrained pleural fluid accumulates in the cavity.



02 / DEPOSITION

Fibrin deposition triggers structural loculation.

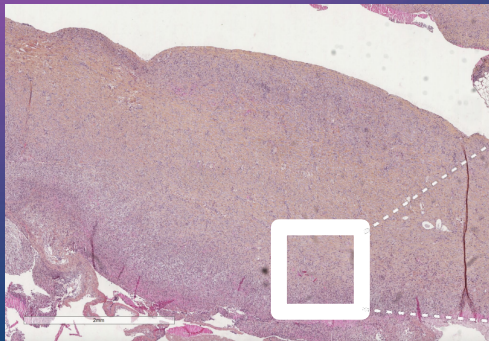


03 / ORGANIZATION

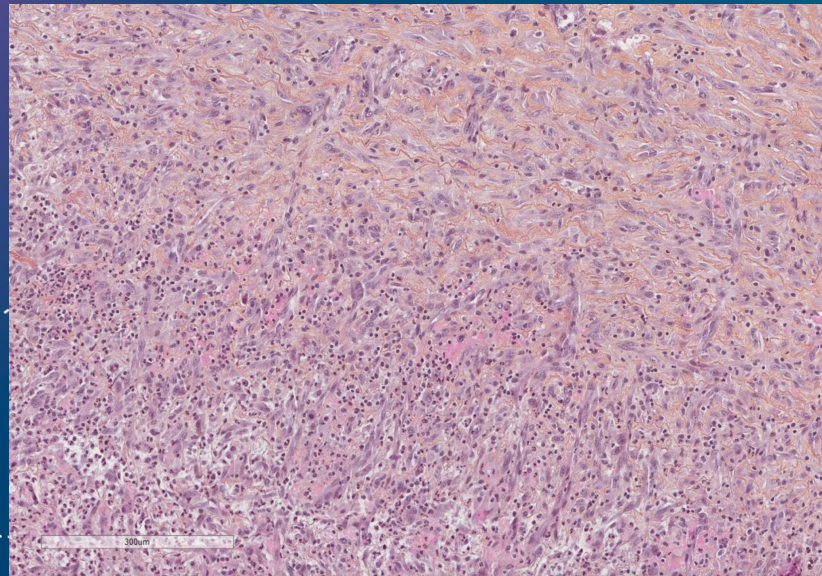
Cross-linked collagen peel fuses to pleural surfaces

The Substrate Mismatch

tPA targets fibrin and DNase targets DNA of pus. Neither drug has activity against collagen.



THE ORGANIZING RIND



Collagenous tissue replacing active inflammation.

Immune cells flood in alongside active new blood vessel growth.

Device Route — Safer Decortication Instrument

HIGH IMPORTANCE, CURRENTLY LOW SATISFACTION

Prolonged air leak (>5 d)

~7.5% today⁷ → **cut 50% relative**

Complete first-pass strip

~39.3% complication rate⁵ → **95% re-expansion, 0 tears**

Plane time on thick rind

112.2 minutes total OR time → **30% faster vs hand**

VATS open conversion

~3-44% today^{9,10} → **<15%**

Biologic Route — Intrapleural Anti-Collagen

HIGH IMPORTANCE, CURRENTLY ZERO SATISFACTION

Re-expansion w/o surgery (lytic-failure)

~5% today → **≥40-50%**

Reduce decortication / referral

~30% reach surgery → **≥30% relative cut**

Option for inoperable patient

zero today → **any re-expansion (FVC<60% → 80%)**

"If there were a way to get rid of the rind without surgery, nobody
would miss doing decortications."

– thoracic surgeon, on the value of a non-surgical option

Biologic Route — Value & Market

Target ASP & Market Size

~\$96

M

Serviceable

Addressable Market
(At \$4,800 ASP)

- 40,000 IPFT/yr¹³ → **50% fail**
- 40% surgery prevented = **8,000 decortifications** avoided
- 8k × \$20k = **\$160M/yr value** at 60% capture

Today's Economics

Payment: ~\$7–11k medical DRG vs ~\$17–31k surgical DRG¹⁴

Thin Margin: a ~15k tpa/DNase course¹² consumes most of the medical DRG

Value: preventing surgery (~\$20k/patient¹¹) avoids the surgical DRG and \$4,800 sits below operation.

Bottom-Up Forecast (ASP \$4.8K, PATIENTS +6.3%/YR)

Year	Procedures	Penetration	Revenue
Yr 1*	20,000	4%	\$3.84M
Yr 5*	25,540	30%	\$36.7M

Path to payment:

Hospital: NTAP (3-yr bridge) then DRG recalibration

Physician: new CPT, starting with Cat III

Drug: NDC → J-code; separate payment if outpatient; inpatient is bundled

*Post 7–10 years spent for Ph 1/2/3 → BLA

¹¹Lacin et al, 2013 | ¹²Kheir et al, 2018 | ¹³Tucker et al, 2016 | ¹⁴CMS FY2025 IPPS Final Rule, MS-DRG relative weights

Path to Market and Key Risks

ENABLERS



Early go/no-go: ≤ 3 months to validate mechanism



Rind space open: The rind has no drug; stripped with tools



Proven enzyme: collagenase already degrades mature collagen

BARRIERS



Time & Capital: Full BLA, no pleural predicate



High Attrition: Drug startup failure rate high



Delivery unproven: Reaching and penetrating rind without lacerating lungs is unsolved

REGULATORY PATHWAY & DEVELOPMENT



Realistic
clinical Path:

Phase I/II: initial
proof in-pleura



Phase II/III:
efficacy + partner



Approval →
Launch

Thank You!

Questions?

MedTech Innovation & Entrepreneurship |

Contact: eraguirr@ncsu.edu

Device Route

TABLE STAKES

Works via existing VATS ports & open / Zero injury
No new capital or added staff / 10–15 min added

BARRIERS TO ADOPTION

Learning curve & practice–pattern change
DRG cost model: must prove concrete savings

REIMBURSEMENT

Uses existing CPT codes; open (32320) & VATS (32652)
Bundled in DRG (caps ASP to hospital savings)

Biologic Route

TABLE STAKES

- Intrapleural via existing chest tube / Bleeding $\leq$ tPA
- Selectivity: no lung/vessel damage / Works regardless

BARRIERS TO ADOPTION

- Full BLA: ~7–10 yr, \$100Ms+, trial-phase failure risk
- MoA unproven / tPA owns fibrin — but tPA fails on the organized collagen rind / Mitigant: VC

REIMBURSEMENT

- NO existing codes; Path: NDC → J-code (multi-yr)
- Precedent: alteplase paid — pathway proven

Device: Value Drivers & Bottom-Up Forecast

VALUE DRIVERS & TARGET PRICING

OR Time Saved: **\$260–780** value ($\$26/\text{min}^{12} \times 10\text{--}30 \text{ min saved}$) — or **\$500–1000** if freed slots convert to added case throughput

Conversion Avoided: **\$347–869** value ($10\text{--}25\% \text{ drop} \times \$3,476 \text{ gap}^{14}$)

Prolonged Air Leak: **~\$96** value ($7.5\% \text{ rate} \times 50\% \text{ cut} \times \$2,556^{13}$)

LOS Reduction: **\$1,500–1,900** value (1 fewer day $\times \sim \$2,000\text{--}2,500$, less $\sim 25\%$ PAL overlap)

Implied ASP: **\$450–650** (indirect levers at 10–30% capture; OR time at 30–40%)

Base ASP below purchasers' WTP (Progel sealant¹⁵ is $\sim \$480/\text{app}$)

SCALABLE AT TARGET ASP

~\$450–650

target ASP

75%

gross margin if
COGS $< \sim \$140$

~\$5–6M

US
decortication-only

BOTTOM-UP FORECAST MODEL

Baseline Assumptions:

$\sim 10,000$ base procedures $\cdot +6\%/\text{yr} \cdot \text{ASP } \$550 \cdot$
S-curve adoption¹⁶ ($3\% \rightarrow 30\%$)

Year 1 — \$0.17M Target

- $10,000 \text{ procedures} \times 3\% \text{ penetration} \times \550

Year 5 — \$2.10M Target

- $12,625 \text{ procedures} \times 30\% \text{ penetration} \times \550

Adhesiolysis: Adjacent Market Opportunity

Atraumatic separation of an adherent fibrous plane off lung — decortication is the beachhead, adhesiolysis is the expansion

COMMON & COSTLY ADHESIONS

Prevalence in Lung Resections

Adhesions are present in
~35–41% of cases^{17, 18}

Severe/Dense Adhesions

The ones needing a specific tool represent **~10–12%** of patients¹⁸

VERIFIED PAIN POINT & SAM

~\$9–18M

Serviceable Addressable Market (At \$550 ASP)

Market grows 2–3x:

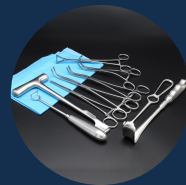
Adhesiolysis (~16–32k cases)²⁰

Independent predictor of conversion, air leak, and longer OR time¹⁹

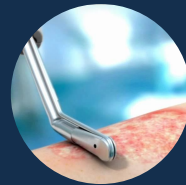
CURRENT CLINICAL PRACTICE



Energy devices



Standard Bipolar Forceps



Hydro-dissection

Path to Market and Key Risks

ENABLERS



FDA Clearance: if device based on existing tech.



Beats Low Bar: Rind is currently stripped with crude tools.



COGS < ~\$140: Molded handpiece off existing pump supports margin.

BARRIERS — COMPETITION / IP



Major Competitors: Reusable tools & large strategics

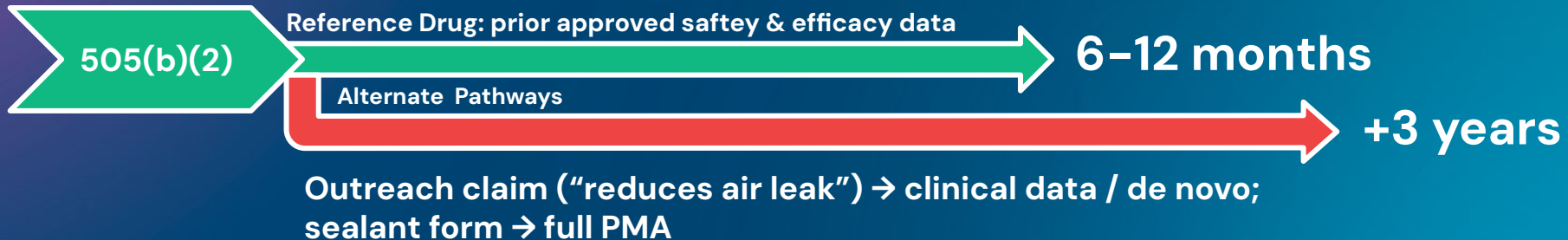


IP Limited : Existing tech patents → design around / differentiate.



Reimbursement: Paid out of the bundled DRG → pure added cost unless savings are proven.

REGULATORY PATHWAY & DEVELOPMENT



Biologic Route — Enablers, Barriers, Path

ENABLERS

Delivery Solved: intrapleural via existing chest tube — no new device required.

505(b)(2) Shortcut: repurpose approved anti-fibrotic (nintedanib / pirfenidone).

High Margins: >85% gross margin post-approval; COGS is a small fraction.

BARRIERS — COMPETITION / IP / ALTERNATIVES

Rind space is open: no approved drug breaks down organized rind; collagenase has never been used in the pleura.

Incumbent: tPA/DNase (Genentech/Roche) owns the fibrin phase, cheap-ish.

Alternatives: decortication surgery, IPC (pleural catheter), and better lytics.

REGULATORY PATHWAY & DEVELOPMENT

Full IND Pathway: Ph 1/2/3 → BLA requires ~7–10 years, \$100Ms+, and has high attrition.

505(b)(2) Accelerant: Only real accelerant.

Clinical Upside: Fast, objective imaging/VC endpoint yields short readout once trials run.

SALES / DISTRIBUTION / SERVICE

Buy-and-Bill Model: J-code, ASP+6% Part B / DRG inpatient; MSLs + formulary drive uptake.

No Field Service: Simple product delivery

Exit Strategy: Out-license to a major respiratory pharma after Ph1/2 proof-of-concept.

Next Steps

VOICE OF CUSTOMER (GENERATIVE)

DEVICE

- Surgeon + value WTP: "what price kills it? what proof do you need?"
- Shadow more decortications to quantify plane-time & air-leak pain

BIOLOGIC

- Pulmonologist interviews: "what would displace a surgical referral?"
- Confirm the inoperable-patient denominator

*Both: NIS procedure-code pull (real volume)
+ pulmonology denominator*

TECHNICAL FEASIBILITY ASSESSMENTS

DEVICE

- Water-jet plane development on cadaver / explanted lung with thick rind
- Measure dissection time + injury / air leak vs hand technique

BIOLOGIC

- Collagen selectivity: remodel rind WITHOUT harming parenchyma or vessels?
- Validate MesoMT / TGF- β / PAI-1 targets in ex vivo pleural tissue

PROTOTYPING & MODELS

DEVICE

- Benchtop atraumatic dissector / nozzle prototypes
- Timed cadaver/porcine decortication; iterate tip & suction

BIOLOGIC

- Source candidate — novel agent or repurposed
- Animal model (tetracycline pleural injury) → re-expansion + selectivity; pre-IND

ESTIMATED INVESTMENT TO REACH THE PIVOT / PERSEVERE DECISION

DEVICE

~\$25–75k · 6–12 months

Buys: prototype + cadaver lab + ~10 surgeon VoC + data pull

BIOLOGIC (proof-of-concept)

~\$1–5M · 12–24 months

Buys: target validation + selectivity + animal PoC + pre-IND. BLA = \$100Ms.